

MySQL_dataset-2

Commit 21:

6340d526b0c738f8de0cbae929e38cf7a06db8ea

Non-C file modifications, therefore no performance regression testing is required.

Commit 22:

218581798af3a003ee397c82e71a82114ba9863e

The 1st symbolic execution

Testcase:CALL GetUserPosts(1);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2.0968
4  The new version is executed by default
5  2.100100000000000003
6  The old version
7  2.0738
8  Comparison of New and old versions (along symbolic execution path)
9  0.011090751277847494
10 Comparison of New and old versions (along the default execution path)
11 0.012682032982930098
```

No performance regression

The 2nd symbolic execution

Testcase:SELECT * FROM products WHERE FIND_IN_SET('New', tags);

No conditional branches, no need for symbolic execution.

```
1  end-to-end diff:
2  The new version executes along symbols
3  7.1431
4  The new version is executed by default
5  7.2003
6  The old version
7  7.1711
8  Comparison of New and old versions (along symbolic execution path)
9  -0.003904561364365365
10 Comparison of New and old versions (along the default execution path)
11 0.004071899708552431
```

No performance regression

The 3rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  83.0404
4  The new version is executed by default
5  84.1876999999999999
6  The old version
7  99.0977
8  Comparison of New and old versions (along symbolic execution path)
9  -0.16203504218564102
10 Comparison of New and old versions (along the default execution path)
11 -0.15045757873290713
```

No performance regression

The 4th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  83.42179999999999
4  The new version is executed by default
5  82.062800000000001
6  The old version
7  84.6717
8  Comparison of New and old versions (along symbolic execution path)
9  -0.014761720858327054
10 Comparison of New and old versions (along the default execution path)
11 -0.030811947793654683
```

No performance regression

The 5th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  83.5207
4  The new version is executed by default
5  82.8498
6  The old version
7  84.4564
8  Comparison of New and old versions (along symbolic execution path)
9  -0.011079089328931816
10 Comparison of New and old versions (along the default execution path)
11 -0.019022833083105605
```

No performance regression

Commit 23:

a4f2714955b56bc27593203bbaf733c934078f0d

Non-C file modifications, therefore no performance regression testing is required.

Commit 24:

1b30b823ad614b320e2eeab97f66daba2e3c2b39

Unable to provide test cases

Commit 25:

6e191c457e48ac4ae897dcaf9720723d49d85f97

Non-C file modifications, therefore no performance regression testing is required.

Commit 26:

da9c26e1885286130309d97306c73dbfc7d24d38

Unable to provide test cases

Commit 27:

5f3490aea8c0f6931906f330f5ad39698915ce03

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  5.0762
4  The new version is executed by default
5  5.0607
6  The old version
7  5.1215
8  Comparison of New and old versions (along symbolic execution path)
9  -0.008845064922386042
10 Comparison of New and old versions (along the default execution path)
11 -0.011871522015034738
```

No performance regression

Commit 28:

ce1137694d2d919b4bac39d45d614ece7443197b

Non-C file modifications, therefore no performance regression testing is required.

Commit 29:

338d9586d2461f6b674a4b7b0125a3ba619b18a8

Non-C file modifications, therefore no performance regression testing is required.

Commit 30:

7c0dda91d700d1de24cf70e4ed9dbc010cb1ebd2

Unable to provide test cases

Commit 31:

167b7afb4d070ec7bc806bcebfcef2e27ab3f2ce

The 1st symbolic execution

Testcase:SELECT * FROM (SELECT user_id FROM posts UNION SELECT user_id FROM comments)
AS derived_table WHERE user_id > 10;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  3.6392
4  The new version is executed by default
5  3.6597999999999997
6  The old version
7  3.6460999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0018924330106139333
10 Comparison of New and old versions (along the default execution path)
11 0.003757439455856956
```

No performance regression

Commit 32:

cb4b9554df2b37a736c1efbdfc182295f02b36c8

Non-C file modifications, therefore no performance regression testing is required.

Commit 33:

0491d9b4d93fb7d49c664196d6f2e97f335e00c4

Unable to provide test cases

Commit 34:

d2c1214b45ac1efc70dcd27bfc737e3bed09a06

The 1st symbolic execution

Testcase:ALTER TABLE users CHANGE COLUMN username username VARCHAR(60);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  3.6713
4  The new version is executed by default
5  3.5987999999999998
6  The old version
7  3.7866
8  Comparison of New and old versions (along symbolic execution path)
9  -0.030449479744361685
10 Comparison of New and old versions (along the default execution path)
11 -0.049595943590556225
```

No performance regression

Commit 35:

645241b5a2317273fa1a78d7a251ae4fcadadb9b

Unable to provide test cases

Commit 36:

3aa93ef9688938ae076afc467339afc57a7ae1a7

Non-C file modifications, therefore no performance regression testing is required.

Commit 37:

46558ef21489ff7209d21abe787ca64370053dce

Unable to provide test cases

Commit 38:

2109aa7ec3a12db3f271fdaf2b7b7dab34cfe761

The 1st symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.00109999999999999998
4  The new version is executed by default
5  4.1028
6  The old version
7  4.028000000000000005
8  Comparison of New and old versions (along symbolic execution path)
9  -0.9997269116186693
10 Comparison of New and old versions (along the default execution path)
11 0.01857000993048653
```

No performance regression

The 2nd symbolic execution

Testcase:ALTER TABLE users ALTER COLUMN email SET DEFAULT 'inactive';

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  3.2531
4  The new version is executed by default
5  3.6753
6  The old version
7  3.7725999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  -0.13770344059799605
10 Comparison of New and old versions (along the default execution path)
11 -0.02579123151142441
```

No performance regression

The 3rd symbolic execution

Testcase:CREATE TABLE IF NOT EXISTS comments_new (id INT AUTO_INCREMENT PRIMARY KEY, post_id INT, comment TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (post_id) REFERENCES posts(id));

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0003000000000000000003
4  The new version is executed by default
5  0.1373
6  The old version
7  0.147
8  Comparison of New and old versions (along symbolic execution path)
9  -0.9979591836734695
10 Comparison of New and old versions (along the default execution path)
11 -0.06598639455782304
```

No performance regression

The 4th symbolic execution

Testcase:UPDATE IGNORE users SET email = 'new_user1@example.com' WHERE id = @user_id;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.001
4  The new version is executed by default
5  0.031
6  The old version
7  0.03210000000000000004
8  Comparison of New and old versions (along symbolic execution path)
9  -0.9688473520249221
10 Comparison of New and old versions (along the default execution path)
11 -0.034267912772585785
```

No performance regression

Commit 39: a7ea07edf2099dfd31026db0979478c62a6f65ab

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  5.1295
```

```
4 The new version is executed by default
5 5.1590000000000001
6 The old version
7 5.2318
8 Comparison of New and old versions (along symbolic execution path)
9 -0.01955349975151948
10 Comparison of New and old versions (along the default execution path)
11 -0.01391490500401374
```

No performance regression

The 2nd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.00120000000000000001
4 The new version is executed by default
5 5.1335
6 The old version
7 5.1781
8 Comparison of New and old versions (along symbolic execution path)
9 -0.9997682547652614
10 Comparison of New and old versions (along the default execution path)
11 -0.00861319789111836
```

No performance regression

Commit 40:

cd66c5cd5ed4d0974a63cbf3e7ab6e8953613777

The 1st symbolic execution

Testcase:INSERT INTO users SET id = 1, username = 'user1', email = 'user1@example.com' ON DUPLICATE KEY UPDATE email = 'updated_user1@example.com';

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0336000000000000005
4  The new version is executed by default
5  0.0422
6  The old version
7  0.0396
8  Comparison of New and old versions (along symbolic execution path)
9  -0.15151515151515146
10 Comparison of New and old versions (along the default execution path)
11 0.06565656565656561
```

There may be performance regression